



Institute of Electronics
National Yang Ming Chiao Tung University
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Midterm Project

Hand-pose estimation (Kaggle)



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Outline

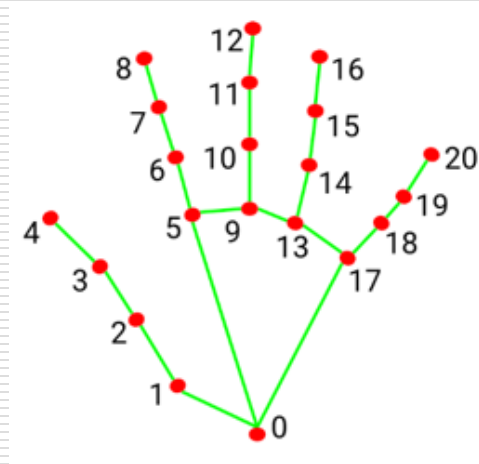
- Midterm project: Hand-pose estimation
 - Task Description
 - Grading and Requirement
 - Submission
 - Kaggle
 - Report
 - Reference
 - Kaggle usage

Task Description : Hand-pose estimation



Task Description: Hand-pose estimation

- The images are collected from FreiHAND dataset
- Dataset provided by TA:
 - Train: 32560 images
 - Test: 1000 images
- DO NOT use the test data, we will regard this as **CHEATING**.



Grading

- Deadline :
 - Kaggle: 8/10 (四) 23:59
- Grading :
 - Kaggle (40%) :
 - › $10 \leq \text{Avg PixelDiff} < 15$ (25%)
 - › $7 \leq \text{Avg PixelDiff} < 10$ (30%)
 - › $5 \leq \text{Avg PixelDiff} < 7$ (35%)
 - › $\text{Avg PixelDiff} < 5$ (40%)
 - Ranking (20% -> 依比例無條件進位)
 - Report (40%)
 - Bonus (10%)

Ranking formula

- Model Size [M-param] $\times$ Cost [G-FLOPs] $\times$ Avg PixelDiff
- Ex: 10 M-param $\times$ 10 G-FLOPs $\times$ 4 = 400
- The smaller the better.

Requirement

- Rules:
 - You can discuss with others, but you should write your homework by yourself!
 - Your codes should be able to generate the results you submit to the leaderboard. (within 1% error)
 - Please build up the model by yourself. (reference some well known model structures are allowed)
 - **DO NOT use the testing data in any way!**

Submission

- Submit your code via email andy.312510138@gmail.com
 - <your English name>_midterm_project.zip
 - Include:
 - > **Your codes (.ipynb)**
 - > **Model weight file (.pth)**
 - > **Report.pdf**
- **DO NOT**
 - Submit the dataset
 - Submit your Project in wrong format
 - > like **.rar/.7z** or **report.doc**

Kaggle

- Register a Kaggle account by yourself
- Join our in-class competition
 - Link: <https://www.kaggle.com/t/2a5f946127d146f68ef55b31fc57cfb3>
 - Maximum daily submission: **10**
- Rules:
 - Your team name should be: < your English name >
> **E.g. Chih-Yao Liang**
> Otherwise, your submission will NOT be graded.
 - You can NOT create multiple Kaggle accounts to submit more results.
 - You can NOT upload results identical to other people's.

Report (40%)

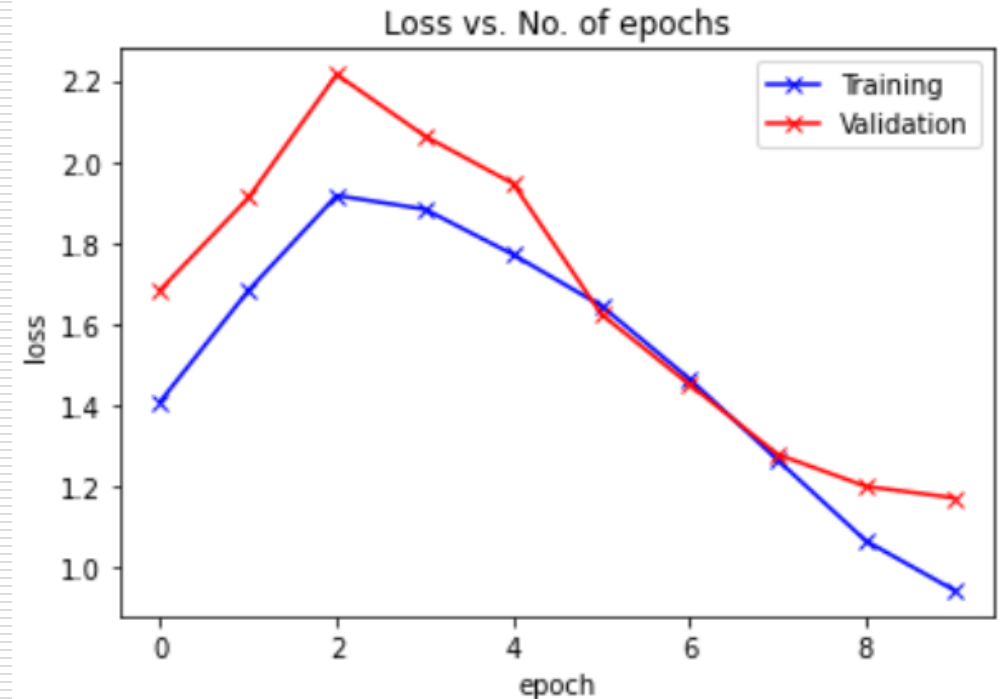
- Model Architecture and Training (20%)
 - Draw your model architecture using simple symbols. (5%)
 - Print learning curve and briefly state your observations. (loss / epoch) (5%)
 - › Hint: you can try some different training methods.
 - What are the differences between training loss and validation loss? (10%)
 - › Hint: plot the train loss and valid loss per epoch together.
 - › Hint: explain the relationship between them.
- How to improve the accuracy (list your methods) (20%):
 - Loss function ?
 - Your network?
 - Activity function?
 - Training skill?
 - Etc...

Report example

Layer (type)	Output Shape	Params	FLOPs (M+A) #
Conv2d-1	[1, 64, 32, 32]	1792	3538944
BatchNorm2d-2	[1, 64, 32, 32]	256	131072
ReLU-3	[1, 64, 32, 32]	0	0
Conv2d-4	[1, 128, 32, 32]	73856	150994944
BatchNorm2d-5	[1, 128, 32, 32]	512	262144
ReLU-6	[1, 128, 32, 32]	0	0
MaxPool2d-7	[1, 128, 16, 16]	0	0
Conv2d-8	[1, 128, 16, 16]	147584	75497472
BatchNorm2d-9	[1, 128, 16, 16]	512	65536
ReLU-10	[1, 128, 16, 16]	0	0
Conv2d-11	[1, 128, 16, 16]	147584	75497472
BatchNorm2d-12	[1, 128, 16, 16]	512	65536
ReLU-13	[1, 128, 16, 16]	0	0
Conv2d-14	[1, 256, 16, 16]	295168	150994944
BatchNorm2d-15	[1, 256, 16, 16]	1024	131072
ReLU-16	[1, 256, 16, 16]	0	0
MaxPool2d-17	[1, 256, 8, 8]	0	0
Conv2d-18	[1, 512, 8, 8]	1180160	150994944
BatchNorm2d-19	[1, 512, 8, 8]	2048	65536
ReLU-20	[1, 512, 8, 8]	0	0
MaxPool2d-21	[1, 512, 4, 4]	0	0
Conv2d-22	[1, 512, 4, 4]	2359808	75497472
BatchNorm2d-23	[1, 512, 4, 4]	2048	16384
ReLU-24	[1, 512, 4, 4]	0	0
Conv2d-25	[1, 512, 4, 4]	2359808	75497472
BatchNorm2d-26	[1, 512, 4, 4]	2048	16384
ReLU-27	[1, 512, 4, 4]	0	0
MaxPool2d-28	[1, 512, 1, 1]	0	0
Flatten-29	[1, 512]	0	0
Dropout-30	[1, 512]	0	0
Linear-31	[1, 100]	51300	102400
ResNet18-32	[1, 100]	0	0

Total parameters: 6,626,020 6.6M
 Trainable parameters: 6,621,540
 Non-trainable parameters: 4,480
 Total flops(M) : 379,684,864 379.7M
 Total flops (M+A): 759,369,728 759.4M

Parameters size (MB): 25.28



Not necessary like this !

Reference

- Colab sample code and video introduction.

<https://drive.google.com/drive/folders/1tpLSH1wd476eeuwY7-tSZKDEey801Va-?usp=sharing>

- 學習使用關鍵字搜尋任何你想問的問題

Kaggle usage

The image displays four screenshots of the Kaggle website interface, illustrating various features and settings.

Top Left Screenshot: Shows the 'ADARLAB_AI_TRAINING' dataset page. The 'Dataset Description' section is visible, along with the 'Files' list (131243 files, 3 GB) and the 'Data Explorer' section. The 'FreiHAND_pub_v2' dataset is highlighted, showing its structure (2 directories, 3 files).

Top Right Screenshot: Shows the 'ADARLAB_AI_TRAINING' dataset page with the 'Dataset Description' section. The 'Files' list (131243 files, 3 GB) and the 'Data Explorer' section are visible. The 'FreiHAND_pub_v2' dataset is highlighted, showing its structure (2 directories, 3 files).

Bottom Left Screenshot: Shows the 'Account' settings page. The 'Your email address' section displays 'andyliang0410@gmail.com'. The 'Your username' section displays 'andyliang0410 (not editable)'. The 'Your account number' section displays '15863872'. The 'Phone verification' section shows 'Verified'. The 'API' section shows 'Create New Token' and 'Expire Token' buttons. The 'Quotas' section shows 'Private Datasets' (916.92 kB / 107.37 GB) and 'GPU' (00:00 / 30 hrs).

Bottom Right Screenshot: Shows the 'Settings' page. The 'API' section shows 'Create New Token' and 'Expire Token' buttons. The 'Your username' section displays 'andyliang0410 (not editable)'. The 'Your account number' section displays '15863872'. A file explorer window is open, showing the 'kaggle.json' file in the 'Downloads' folder.

Thank you